GOOGLE PUBLIC SECTOR HACKATHON SUBMISSION

Fairfax County Resource Navigator

A Zero-PII semantic routing prototype that turns natural-language resident needs into source-grounded Fairfax County service recommendations.

MAJOR

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Purpose & Vision

Residents often know what they need, but not what Fairfax County calls it, which department owns it, or which portal contains the right next step. Fragmented discovery across PDFs, department pages, and program sites can create service deserts for residents already facing time, language, mobility, disability, or digital-access barriers.

The Resource Navigator reframes discovery around intent. Residents ask in plain language, share only safe context such as ZIP code, age, household size, or interests, and receive an official-source route. The result supports One Fairfax by making the first step toward help faster, clearer, and more equitable.

How The Prototy

1. Streamlit-style intake ca non-PII context.
2. A local SentenceTransf into a semantic vector.
3. A Pandas Shadow Data tags, age/ZIP hints, and
4. Guardrails route emerg avoid eligibility promise
5. The UI returns the depa steps, and source citati

Zero PII

No full names, exact addresses, SSNs, account numbers, or resident records required.

8 Domains

Housing, taxes, health, parks, permits, utilities, transportation, and safety.

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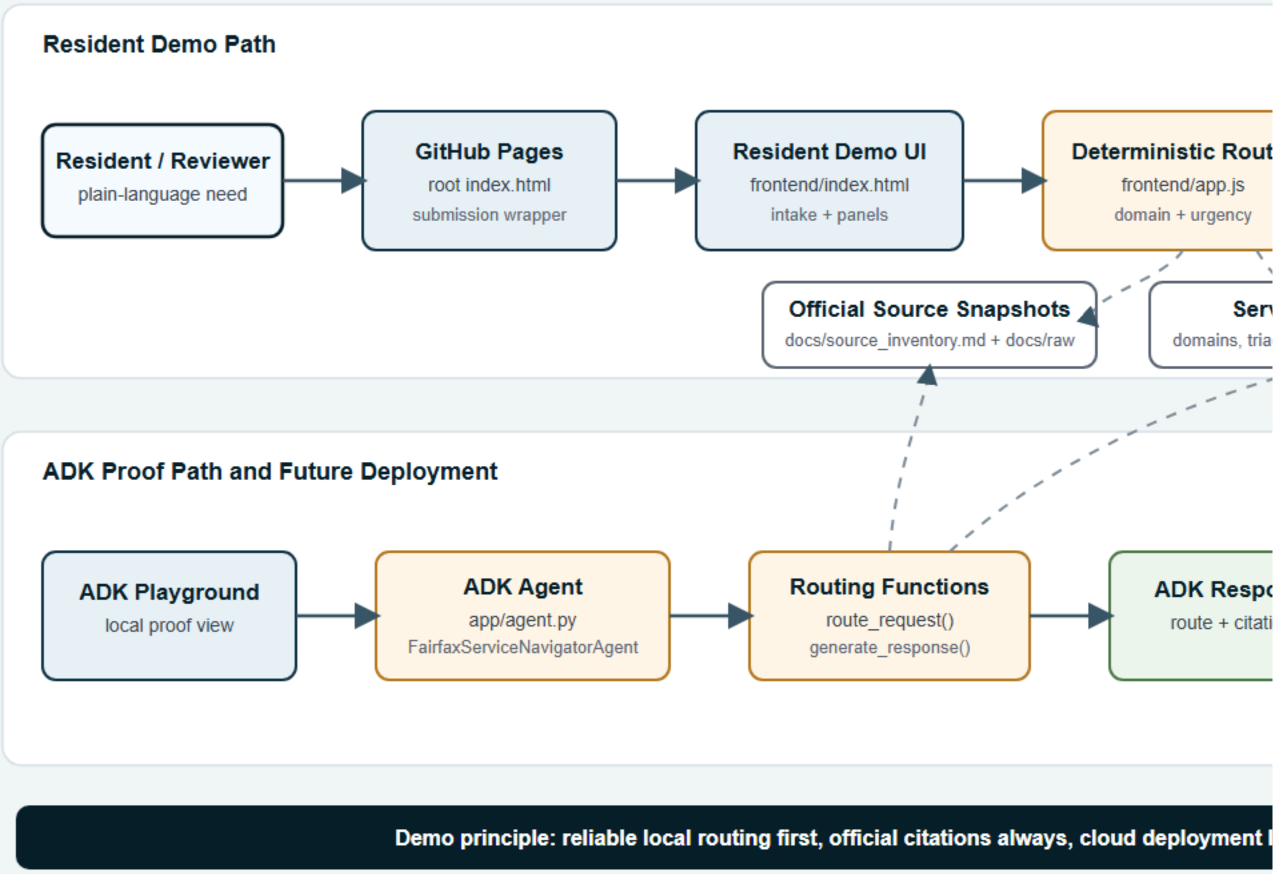
State
Goog

Architecture Flow

Fairfax Resource Navigator Architecture

Local static demo plus ADK proof path, grounded in official Fairfax source snapshots



Housing & CSP	**Taxes & DTA**	**Health, CSB & NCS**
Permits & Code	**Public Works**	**Transportation**

Why it wins: rapid deployment, privacy by design, official-source grounding, and immediate Fairfax equity goals.

Rapid Deployment

Runs as a lightweight Python app for demos and can containerize cleanly for Cloud Run without a complex infrastructure footprint.

Privacy By Design

The Shadow Database is service metadata, not resident data. Matching uses safe demographic proxies only when they improve routing.

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Help
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Demo status: mock-first local prototype; no live model API required for chat.

Submission assets: README, one-pager, archi